

W2 – 4.4 Compound Angle Formulas

MHF4U

Jensen

1) Use an appropriate compound angle formula to express as a single trig function, and then determine an exact value for each

a) $\sin \frac{\pi}{4} \cos \frac{\pi}{12} + \cos \frac{\pi}{4} \sin \frac{\pi}{12}$

$$\begin{aligned} &= \sin \left(\frac{\pi}{4} + \frac{\pi}{12} \right) = \sin \left(\frac{\pi}{3} \right) \\ &= \sin \left(\frac{3\pi}{12} + \frac{\pi}{12} \right) = \frac{\sqrt{3}}{2} \\ &= \sin \left(\frac{4\pi}{12} \right) \end{aligned}$$

b) $\sin \frac{\pi}{4} \cos \frac{\pi}{12} - \cos \frac{\pi}{4} \sin \frac{\pi}{12}$

$$\begin{aligned} &= \sin \left(\frac{\pi}{4} - \frac{\pi}{12} \right) = \sin \left(\frac{\pi}{6} \right) \\ &= \sin \left(\frac{3\pi}{12} - \frac{\pi}{12} \right) = \frac{1}{2} \\ &= \sin \left(\frac{2\pi}{12} \right) \end{aligned}$$

c) $\cos \frac{\pi}{4} \cos \frac{\pi}{12} - \sin \frac{\pi}{4} \sin \frac{\pi}{12}$

$$\begin{aligned} &= \cos \left(\frac{\pi}{4} + \frac{\pi}{12} \right) = \cos \left(\frac{\pi}{3} \right) \\ &= \cos \left(\frac{3\pi}{12} + \frac{\pi}{12} \right) = \frac{1}{2} \\ &= \cos \left(\frac{4\pi}{12} \right) \end{aligned}$$

d) $\cos \frac{\pi}{4} \cos \frac{\pi}{12} + \sin \frac{\pi}{4} \sin \frac{\pi}{12}$

$$\begin{aligned} &= \cos \left(\frac{\pi}{4} - \frac{\pi}{12} \right) = \cos \left(\frac{\pi}{6} \right) \\ &= \cos \left(\frac{3\pi}{12} - \frac{\pi}{12} \right) = \frac{\sqrt{3}}{2} \\ &= \cos \left(\frac{2\pi}{12} \right) \end{aligned}$$

e) $\cos \frac{2\pi}{9} \cos \frac{5\pi}{18} - \sin \frac{2\pi}{9} \sin \frac{5\pi}{18}$

$$\begin{aligned} &= \cos \left(\frac{2\pi}{9} + \frac{5\pi}{18} \right) = \cos \left(\frac{\pi}{2} \right) \\ &= \cos \left(\frac{4\pi}{18} + \frac{5\pi}{18} \right) = 0 \\ &= \cos \left(\frac{9\pi}{18} \right) \end{aligned}$$

f) $\cos \frac{10\pi}{9} \cos \frac{5\pi}{18} + \sin \frac{10\pi}{9} \sin \frac{5\pi}{18}$

$$\begin{aligned} &= \cos \left(\frac{10\pi}{9} - \frac{5\pi}{18} \right) = \cos \left(\frac{5\pi}{6} \right) \\ &= \cos \left(\frac{20\pi}{18} - \frac{5\pi}{18} \right) = -\cos \left(\frac{\pi}{6} \right) \\ &= \cos \left(\frac{15\pi}{18} \right) = -\frac{\sqrt{3}}{2} \end{aligned}$$

3) Apply a compound angle formula, and then determine an exact value for each.

a) $\sin \left(\frac{\pi}{3} + \frac{\pi}{4} \right)$

$$\begin{aligned} &= \sin \frac{\pi}{3} \cos \frac{\pi}{4} + \cos \frac{\pi}{3} \sin \frac{\pi}{4} \\ &= \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) + \left(\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) \\ &= \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} \\ &= \frac{\sqrt{6} + \sqrt{2}}{4} \end{aligned}$$

b) $\cos \left(\frac{\pi}{3} + \frac{\pi}{4} \right)$

$$\begin{aligned} &= \cos \frac{\pi}{3} \cos \frac{\pi}{4} - \sin \frac{\pi}{3} \sin \frac{\pi}{4} \\ &= \left(\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) - \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) \\ &= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} \\ &= \frac{\sqrt{2} - \sqrt{6}}{4} \end{aligned}$$

c) $\cos \left(\frac{2\pi}{3} - \frac{\pi}{4} \right)$

$$\begin{aligned} &= \cos \frac{2\pi}{3} \cos \frac{\pi}{4} + \sin \frac{2\pi}{3} \sin \frac{\pi}{4} \\ &= -\cos \frac{\pi}{3} \cos \frac{\pi}{4} + \sin \frac{\pi}{3} \sin \frac{\pi}{4} \\ &= \left(-\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) + \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) \\ &= -\frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4} \\ &= \frac{-\sqrt{2} + \sqrt{6}}{4} \end{aligned}$$

d) $\sin \left(\frac{2\pi}{3} - \frac{\pi}{4} \right)$

$$\begin{aligned} &= \sin \frac{2\pi}{3} \cos \frac{\pi}{4} - \cos \frac{2\pi}{3} \sin \frac{\pi}{4} \\ &= \sin \frac{\pi}{3} \cos \frac{\pi}{4} - (-\cos \frac{\pi}{3}) \sin \frac{\pi}{4} \\ &= \left(\frac{\sqrt{3}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) + \left(\frac{1}{2} \right) \left(\frac{\sqrt{2}}{2} \right) \\ &= \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} \\ &= \frac{\sqrt{6} + \sqrt{2}}{4} \end{aligned}$$

$$\text{e) } \tan\left(\frac{\pi}{4} + \pi\right)$$

$$= \frac{\tan \frac{\pi}{4} + \tan \pi}{1 - \tan \frac{\pi}{4} \tan \pi}$$

$$= \frac{1 + 0}{1 - 1(0)}$$

$$= \frac{1}{1}$$

$$= 1$$

$$\text{f) } \tan\left(\frac{\pi}{3} - \frac{\pi}{6}\right)$$

$$= \frac{\tan \frac{\pi}{3} - \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{3} \tan \frac{\pi}{6}}$$

$$= \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1 + \sqrt{3} \left(\frac{1}{\sqrt{3}}\right)}$$

$$= \frac{\frac{3}{\sqrt{3}} - \frac{1}{\sqrt{3}}}{2}$$

$$= \frac{\left(\frac{2}{\sqrt{3}}\right)}{\left(\frac{2}{1}\right)}$$

$$= \frac{2}{\sqrt{3}} \times \frac{1}{2}$$

$$= \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{\sqrt{3}}{3}$$

4) Use an appropriate compound angle formula to determine an exact value for each.

$$\text{a) } \sin \frac{7\pi}{12} = \sin\left(\frac{3\pi}{12} + \frac{4\pi}{12}\right)$$

$$= \sin\left(\frac{\pi}{4} + \frac{\pi}{3}\right)$$

$$= \sin \frac{\pi}{4} \cos \frac{\pi}{3} + \cos \frac{\pi}{4} \sin \frac{\pi}{3}$$

$$= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}$$

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$\text{b) } \sin \frac{5\pi}{12} = \sin\left(\frac{2\pi}{12} + \frac{3\pi}{12}\right)$$

$$= \sin\left(\frac{\pi}{6} + \frac{\pi}{4}\right)$$

$$= \sin \frac{\pi}{6} \cos \frac{\pi}{4} + \cos \frac{\pi}{6} \sin \frac{\pi}{4}$$

$$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}$$

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$

$$\text{c) } \cos \frac{11\pi}{12}$$

$$= \cos\left(\frac{8\pi}{12} + \frac{3\pi}{12}\right)$$

$$= \cos\left(\frac{2\pi}{3} + \frac{\pi}{4}\right)$$

$$= \cos \frac{2\pi}{3} \cos \frac{\pi}{4} - \sin \frac{2\pi}{3} \sin \frac{\pi}{4}$$

$$= -\cos \frac{\pi}{3} \cos \frac{\pi}{4} - \sin \frac{\pi}{3} \sin \frac{\pi}{4}$$

$$= \left(-\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

$$= -\frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

$$= -\frac{\sqrt{2} + \sqrt{6}}{4}$$

$$\text{d) } \cos \frac{5\pi}{12} = \cos\left(\frac{2\pi}{12} + \frac{3\pi}{12}\right)$$

$$= \cos\left(\frac{\pi}{6} + \frac{\pi}{4}\right)$$

$$= \cos \frac{\pi}{6} \cos \frac{\pi}{4} - \sin \frac{\pi}{6} \sin \frac{\pi}{4}$$

$$= \frac{\sqrt{3}}{2} \left(\frac{\sqrt{2}}{2}\right) - \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

$$= \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\text{e) } \sin \frac{13\pi}{12} = \sin\left(\frac{9\pi}{12} + \frac{4\pi}{12}\right)$$

$$= \sin\left(\frac{3\pi}{4} + \frac{\pi}{3}\right)$$

$$= \sin \frac{3\pi}{4} \cos \frac{\pi}{3} + \cos \frac{3\pi}{4} \sin \frac{\pi}{3}$$

$$= \sin \frac{\pi}{4} \cos \frac{\pi}{3} + (-\cos \frac{\pi}{4}) \left(\sin \frac{\pi}{3}\right)$$

$$= \frac{\sqrt{2}}{2} \left(\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

$$\text{f) } \cos \frac{17\pi}{12} = \cos\left(\frac{15\pi}{12} + \frac{2\pi}{12}\right)$$

$$= \cos\left(\frac{5\pi}{4} + \frac{\pi}{6}\right)$$

$$= \cos \frac{5\pi}{4} \cos \frac{\pi}{6} - \sin \frac{5\pi}{4} \sin \frac{\pi}{6}$$

$$= -\cos \frac{\pi}{4} \cos \frac{\pi}{6} - (-\sin \frac{\pi}{4}) \sin \frac{\pi}{6}$$

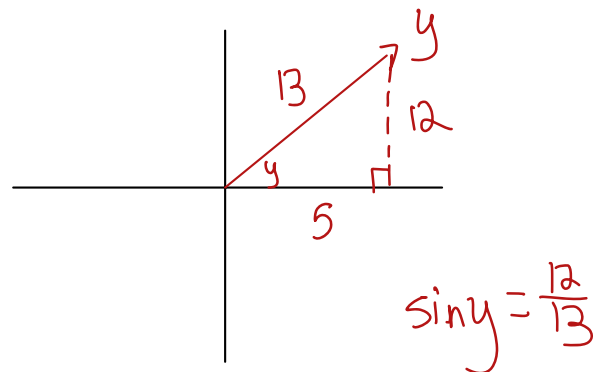
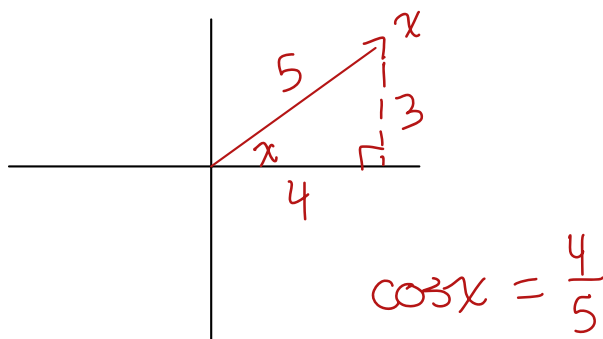
$$= \left(-\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right) \left(\frac{1}{2}\right)$$

$$= -\frac{\sqrt{6} + \sqrt{2}}{4}$$

$$\begin{aligned}
 \text{g) } \sin \frac{19\pi}{12} &= \sin \left(\frac{15\pi}{12} + \frac{4\pi}{12} \right) \\
 &= \sin \left(\frac{5\pi}{4} + \frac{\pi}{3} \right) \\
 &= \sin \frac{5\pi}{4} \cos \frac{\pi}{3} + \cos \frac{5\pi}{4} \sin \frac{\pi}{3} \\
 &= -\sin \frac{\pi}{4} \cos \frac{\pi}{3} + (-\cos \frac{\pi}{4}) \left(\sin \frac{\pi}{3} \right) \\
 &= \left(-\frac{\sqrt{2}}{2} \right) \left(\frac{1}{2} \right) - \left(\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) \\
 &= \frac{-\sqrt{2} - \sqrt{6}}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{h) } \cos \frac{23\pi}{12} &= \cos \left(\frac{15\pi}{12} + \frac{8\pi}{12} \right) \\
 &= \cos \left(\frac{5\pi}{4} + \frac{2\pi}{3} \right) \\
 &= \cos \frac{5\pi}{4} \cos \frac{2\pi}{3} - \sin \frac{5\pi}{4} \sin \frac{2\pi}{3} \\
 &= -\cos \frac{\pi}{4} \left(-\cos \frac{\pi}{3} \right) - \left(-\sin \frac{\pi}{4} \right) \left(\sin \frac{\pi}{3} \right) \\
 &= \left(-\frac{\sqrt{2}}{2} \right) \left(-\frac{1}{2} \right) + \left(\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) \\
 &= \frac{\sqrt{2} + \sqrt{6}}{4}
 \end{aligned}$$

5) Angles x and y are located in the first quadrant such that $\sin x = \frac{3}{5}$ and $\cos y = \frac{5}{13}$. Determine exact values for $\cos x$ and $\sin y$.



6) Refer to the previous question. Determine an exact value for each of the following.

a) $\sin(x + y)$

$$\begin{aligned}
 &= \sin x \cos y + \cos x \sin y \\
 &= \left(\frac{3}{5} \right) \left(\frac{5}{13} \right) + \left(\frac{4}{5} \right) \left(\frac{12}{13} \right) \\
 &= \frac{15}{65} + \frac{48}{65} \\
 &= \frac{63}{65}
 \end{aligned}$$

b) $\sin(x - y)$

$$\begin{aligned}
 &= \sin x \cos y - \cos x \sin y \\
 &= \left(\frac{3}{5} \right) \left(\frac{5}{13} \right) - \left(\frac{4}{5} \right) \left(\frac{12}{13} \right) \\
 &= \frac{15}{65} - \frac{48}{65} \\
 &= \frac{-33}{65}
 \end{aligned}$$

c) $\cos(x + y)$

$$\begin{aligned}
 &= \cos x \cos y - \sin x \sin y \\
 &= \left(\frac{4}{5} \right) \left(\frac{5}{13} \right) - \left(\frac{3}{5} \right) \left(\frac{12}{13} \right) \\
 &= \frac{20}{65} - \frac{36}{65} \\
 &= \frac{-16}{65}
 \end{aligned}$$

d) $\cos(x - y)$

$$\begin{aligned}
 &= \cos x \cos y + \sin x \sin y \\
 &= \left(\frac{4}{5} \right) \left(\frac{5}{13} \right) + \left(\frac{3}{5} \right) \left(\frac{12}{13} \right) \\
 &= \frac{20}{65} + \frac{36}{65} \\
 &= \frac{56}{65}
 \end{aligned}$$

7) Use a compound angle formula to show that $\cos(2x) = \cos^2 x - \sin^2 x$

LS

$$= \cos(2x)$$

$$= \cos(x + x)$$

$$= \cos x \cos x - \sin x \sin x$$

$$= \cos^2 x - \sin^2 x$$

RS

$$= \cos^2 x - \sin^2 x$$

LS=RS